Sachin Saailesh Jeyakkumaran

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Summary

AI Engineer with 3+ years production experience in machine learning (ML), computer vision, NLP. Expert in scalable software development, algorithms, PyTorch, TensorFlow, distributed systems achieving 99.2% accuracy, 87% predictive maintenance. Proven debugging, statistical analysis, cross-functional leadership, Docker/Kubernetes, delivering production AI solutions across healthcare, manufacturing, cybersecurity.

Technical Skills

Programming Languages: Python, C++, C, SQL, Java, Bash scripting
Machine Learning (ML) & Deep Learning: Machine Learning, Deep Learning, PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Computer Vision (YOLO, RT-DETR, FaceNet, ResNet, EfficientNet, 3D CNNs), Natural Language Processing (NLP), BERT, Transformers, Large Language Models (LLMs), Time-Series Forecasting (LSTM, SARIMA, Prophet), Reinforcement Learning (RL), Anomaly Detection
Computer Vision & Image Processing: Object Detection, Face Recognition, Activity Recognition, Defect Detection, Image Processing, Pose Estimation, Person Re-Identification, Temporal Action Localization, OpenCV, MediaPipe
Software Engineering & MLOps: Software Development, Software Engineering, Algorithms, Data Structures, Scalability, Distributed Computing, Debugging, Testing, Quality Assurance (QA), Docker, Kubernetes, MLflow, TensorFlow Lite, ONNX, TensorRT, TorchScript, Model Quantization (INT8/FP16), Edge AI (NVIDIA Jetson), CI/CD pipelines, Model Monitoring, A/B Testing
Production AI Systems: FastAPI, REST APIs, Real-time Inference Pipelines, Data Processing, Multi-Stream Processing, Predictive Maintenance, Feature Engineering, Intrusion Detection, Explainable AI (Grad-CAM, SHAP), Software Systems, Prometheus
Data Science & Analytics: Statistical Analysis, Pandas, NumPy, Apache Spark, Time-Series Analysis, Grafana, Tableau, Power BI, MySQL
Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker), Azure, Google Cloud Platform, Redis, Kafka, Linux, Git, Vector Databases (Qdrant)

Education

State University of New York at Buffalo, Master of Science (MS), Artificial Intelligence - *Buffalo NY* — *2024 – 2026*

Sri Ramachandra Faculty of Engineering and Technology, Bachelor of Technology (B.Tech.), Computer Science & Engineering (AI & ML) - *Chennai, India* — *2019 – 2023*

Experience

Pisence Technologies, **AI Engineer** - *Chennai, India* — *Aug 2023 – Jul 2024*

- Developed scalable ML system using algorithms (YOLOv8, FaceNet) achieving 99.2% accuracy, processing 100+ streams <150ms; built activity recognition with 3D CNNs on edge devices at 30 FPS, collaborating with cross-functional team.
- Engineered ML models using statistical analysis (XGBoost, LSTM) achieving 87% prediction 24-48 hours advance, reduced downtime 28%; designed distributed forecasting pipeline processing 10M+ readings daily, improving OEE 18% through research.
- Implemented defect detection software (ResNet, EfficientNet) achieving 94% accuracy on 5TB+ imagery with debugging, QA; developed energy forecasting using algorithms (SARIMA) <8% MAPE, achieving 22% savings, 27% waste reduction.
- Built ML intrusion detection using algorithms (Isolation Forest, Autoencoders) 96% accuracy <2% false positives on 50M+ logs; developed NLP phishing detection (BERT) 98% accuracy, reduced time 45 to 8 minutes, mentoring engineers.

Calidad Healthcare, **AI/ML Engineer** - *Chennai, India* — *Jun 2022 – Jul 2023*

- Architected production ML pipeline for 4 imaging modalities using software engineering; implemented labeling with k-fold validation, reducing errors 35%, achieving 95% AUC, F1 0.94, mAP 0.58 through applied research, statistical analysis.
- Engineered 5 deep learning architectures combining CNN/transformers in PyTorch, TensorFlow; optimized with focal loss, class balancing with debugging, QA, demonstrating problem-solving in research, cross-functional team.
- Deployed scalable MLOps using Docker, FastAPI, MLflow, CI/CD, A/B testing; serving 10K+ predictions 99.5% uptime, providing technical leadership, mentorship to teams, implementing software development best practices, improvement.
- Optimized ML models for edge deployment via quantization, TFLite using algorithms for scalability, reducing inference 60%, size 4x; integrated Grad-CAM/SHAP for explainable AI, compliance, supporting data-driven decision making, reliability.

Sri Ramachandra Institute of Higher Education and Research, **Teaching Assistant** - *Chennai, India* — *Feb 2023 – Jun 2023*

- Led technical instruction for 120+ students in Data Visualization, Computer Networks, Ethical Hacking; designed 15+ hands-on labs using Pandas, NumPy, Tableau, Power BI, demonstrating leadership, communication, cross-functional collaboration.

Projects

Production RAG System: AI-Powered Code Intelligence with LLMs — *Jun 2025 – Sep 2025*

- Built enterprise ML code intelligence using LLMs, Retrieval-Augmented Generation (RAG), hybrid retrieval (Qdrant + BM25 + Cohere); improved precision 22% with sub-12s latency for 50K+ codebases through distributed processing, and applied research.
- Deployed modular AI analyzers for security scanning via FastAPI REST APIs, Gradio, Prometheus; optimized to 3.4 GB memory, reduced hallucinations 29%, increased throughput 25% using Agile software development, debugging, quality assurance practices.

Real-Time Weapon Detection AI: Computer Vision Benchmark Study — *Feb 2025 – May 2025*

- Benchmarked YOLOv11, RT-DETR machine learning models for AI-powered CCTV weapon detection using algorithms; trained with mixed-precision on 5,149 images, optimizing augmentation, anchor tuning with software engineering skills in Python, TensorFlow.
- Achieved RT-DETR mAP50 83.0 / mAP50–95 37.2 (18.7ms); YOLOv11 mAP50 77.8 / 34.9 (3.1ms). Built COCO evaluation toolkit; exported TorchScript/ONNX for production with documentation, demonstrating research and statistical analysis.